Day 2 Handout: Randomized Experiments and A/B Testing

Experimentation and Causal Inference, SDAIA Academy. Revision notes for participants.

What Day 2 was about

Day 1 showed that randomization buys a causal claim. Day 2 turned that principle into an operational experiment: choosing the randomization unit and scheme, sizing the test before launch, estimating with a design-matched standard error, sharpening with pre-treatment covariates, and running the test in production without the failure modes that turn confident wins into losses.

Anatomy of an experiment

Unit of randomization: what gets independently assigned (user, session, region). It must match the level of interference and of the decision.

Arms: control plus one or more variants. More arms split power and multiply comparisons.

Randomization scheme: the probabilistic rule mapping units to arms; the source of all causal validity.

Primary metric (OEC) and guardrails: exactly one pre-specified decision metric, plus metrics that must not degrade (latency, errors, complaints).

The Injaz uploader design: unit = user (national-ID hash); OEC = service-completion rate; guardrails = time-to-complete, support-ticket rate, upload-error rate; trigger = reaching the upload step; unit of analysis = user.

Randomization schemes

Scheme	What it buys	What it costs
Complete	Simplicity	Chance imbalance at small n
Block or stratified	Balance by design; lower variance (roughly by the share of outcome variance the blocking variable explains)	Analysis must include stratum indicators
Cluster	Contains interference; matches group-level delivery	Design effect 1 + (m minus 1) rho; need more clusters, must cluster SEs
Factorial	Main effects and interactions in one test	Pre-specify interactions

Randomize at the coarsest unit that interference demands, and no coarser. Randomize by user rather than session when users return and must see a consistent experience. Never analyze finer than you assigned without clustering.

Assignment mechanics: hash salt plus unit id to a number in [0, 1) and bucket it. Deterministic, stateless, reproducible, and independent across concurrent experiments if each has its own salt.

Sample-ratio mismatch (SRM): a chi-square test of observed arm counts against the intended split. A failure almost always signals a bug in assignment or logging. Never analyze an experiment that fails SRM.

Inference for experiments

Randomization inference (Fisher): pick the statistic before seeing data; permute labels respecting the design; the p-value is the share of permutations at least as extreme. Exact and assumption-free.

Neyman: tau-hat = mean(Y1) minus mean(Y0), unbiased for the ATE; variance S1 squared over n1 plus S0 squared over n0 (conservative). Under complete randomization this equals the HC2 robust SE from regressing Y on T.

Fisher tests a sharp null and gives no effect size; Neyman gives an estimate and interval. Use Fisher for small samples and clean significance, Neyman when you need an effect size.

Reading a result honestly: a result is a triple of point estimate in real units, interval, and assumptions ledger (unit, SE type, adjustment). Statistical and practical significance differ.

p-values: p is P(data at least this extreme given the null), not P(null given data). p > 0.05 is not evidence of no effect. A tiny p on a huge sample can accompany a trivial effect. The interval usually carries more decision information.

Regression, CUPED and heterogeneity

Regression estimator: Y = a + tau T + e gives the difference in means; use robust SEs. Adding pre-treatment covariates (Y = a + tau T + X b + e) leaves tau unbiased and usually reduces variance. Lin’s estimator interacts T with centered covariates to guarantee no efficiency loss. Adjust only for pre-treatment variables.

CUPED: regress the outcome on a strongly correlated pre-period metric and analyze the residual. Y_adj = Y minus theta (X_pre minus mean X_pre), with theta = cov(Y, X_pre) / var(X_pre). Variance falls by roughly the squared pre/post correlation; a correlation of 0.7 halves variance, the same as doubling the sample. Unbiased because the covariate is pre-treatment.

Heterogeneous effects (CATE): the ATE hides winners and losers. Meta-learners: S (one model, treatment as a feature), T (one model per arm), X (imputed effects plus propensity weighting, for unequal arms), DR/R (doubly robust, residual-based). Pre-register a small set of subgroup hypotheses; treat the rest as exploratory and correct for multiplicity; validate promising CATEs in a fresh test before acting.

A/B testing in production

Lifecycle: hypothesis and metric; power and sample size; randomize and instrument (A/A test, SRM); analyze and decide at the pre-set horizon.

Power quartet: alpha (false-positive rate, usually 0.05), power 1 minus beta (usually 0.80), minimum detectable effect (a business decision), sample size n. Fix three and the fourth follows.

Sample size per arm for proportions:

n = (z_alpha/2 + z_beta)^2 [p1(1 minus p1) + p2(1 minus p2)] / (p2 minus p1)^2

For means: n = 2 sigma^2 (z_alpha/2 + z_beta)^2 / delta^2. With alpha 0.05 and power 0.80, z_alpha/2 + z_beta is about 2.80. Halving the MDE quadruples n. Worked examples: baseline 0.55, MDE 2 points gives about 9,700 per arm; baseline 0.10, MDE 0.5 points gives about 56,400 per arm.

Run length = n per arm times number of arms, divided by (daily eligible traffic times trigger rate), with a floor of one to two full weeks to average over weekly seasonality and novelty effects.

MDE at fixed n: MDE is about (z_alpha/2 + z_beta) times sqrt(2 p(1 minus p) / n).

Peeking: a fixed-horizon p-value is valid once, at the planned n. Checking daily and stopping at the first p < 0.05 pushes the true false-positive rate from 5 percent to 20 to 30 percent (Lab 2: about 5, 13, 20, 25 percent at 1, 5, 10, 20 looks). If you want to watch continuously you must use a method built for it: group-sequential alpha spending, SPRT, always-valid confidence sequences, or pre-specified Bayesian monitoring.

Multiple testing: 20 tests at alpha 0.05 under the null give a 64 percent chance of at least one false positive. Bonferroni controls the family-wise error rate (use for the guardrail family); Benjamini-Hochberg controls the false discovery rate (use for exploratory metric and segment sweeps). One pre-registered OEC needs no correction. A “significant” segment from an unplanned sweep is a hypothesis for the next test, not a finding to ship.

Failure modes: sample-ratio mismatch, peeking, interference, novelty and primacy effects, Simpson’s paradox, winner’s curse (the shipped variant’s true effect is smaller than the estimate that got it shipped).

Lab 2 recap

Sizing: 9,666 per arm by the closed form against 9,669 from statsmodels; a two-point lift was easily detectable at Injaz traffic. Integrity: assignment reproduced from the hash, SRM passed on clean data and failed when 3 percent of one arm was dropped, all SMDs below 0.01. A/A test: about 5 percent false positives over 500 replications. Effect: +2.2 points, two-proportion equal to OLS; the pre-period covariate cut variance by about 13 percent. Guardrails clean under Bonferroni; five of 31 segments raw-significant, none survived BH. Peeking inflated the error rate to about 25 percent at 20 looks; the mSPRT boundary brought it back under 5 percent while keeping full power.

Exercise: audit this experiment report (Hour 5 discussion)

“We tested a redesigned dashboard for returning Injaz users. Sessions were randomized 50/50 into old and new dashboard. We tracked five primary metrics: completion rate, time on page, clicks, return visits within 7 days, and satisfaction. We checked results every morning and stopped on day 6 when completion rate reached $p = 0.04$. Standard errors are from a standard t-test on sessions. Although the overall effect on return visits was flat, we found a significant lift among iOS users in Riyadh aged 25 to 34 ($p = 0.02$) and recommend targeting that segment.”

Find at least five flaws using: randomization unit versus interference; horizon and peeking; number of primary metrics and guardrails; standard errors versus design; pre-registered versus fished subgroups.

Self-check (answers at the end)

1. A campaign is delivered per city and neighbors may hear about it. Best design?
2. Analyzing a cluster-randomized test at the individual level will typically do what to the standard error?
3. Under complete randomization, the Neyman SE for the difference in means corresponds to which regression SE?
4. Halving the MDE changes required sample size by roughly what factor?
5. The arms come out 55/45 instead of 50/50. What is this and what do you do?
6. Why must CUPED’s covariate be pre-treatment?
7. Bonferroni versus Benjamini-Hochberg: which controls what, and which family gets which?
8. “$p = 0.20$, so the treatment has no effect.” Correct or not?

Answers

1. Cluster randomization by city.
2. Understate it (false precision).
3. HC2 robust.
4. About four times (n is proportional to 1/MDE squared).
5. Sample-ratio mismatch; do not analyze until the assignment or logging bug is found.
6. So treatment cannot affect it; the adjustment then removes noise without bias.
7. Bonferroni controls the family-wise error rate (guardrails); BH controls the false discovery rate (exploratory sweeps).
8. Incorrect; it confuses absence of evidence with evidence of absence.

Audit exercise flaws: session-level randomization for returning users (contamination, wrong unit); five primary metrics with no guardrails; daily peeking with a fixed-horizon test and stopping on the first green; session-level t-test ignoring within-user correlation; a fished three-way subgroup presented as a finding; no power calculation or pre-set horizon.